CUAD

Cornell University Autonomous Drone

2025-2026 Student Project Team

2023-2024 SPONSORSHIP PACKET

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ABOUT US

FOUNDED IN 2017, WE ARE A VIBRANT AND DEDICATED TEAM COMMITTED TO ADVANCING THE FIELD OF AERIAL ROBOTICS. SPECIALIZING IN BOTH CONTROLLED AND AUTONOMOUS DRONES, WE NOW STRIVE TO PUSH THE LIMITS OF SPEED AND FLIGHT CAPABILITIES.

This year, we are developing multiple high-performance drones, including a missile-style autonomous glider designed to be released from our large carrier drone, deploy its wings, and operate as a surveillance platform. We are also creating a gesture-controlled drone that responds to user movements, and upgrading our unique drone, Buttery, to autonomously navigate an obstacle course and retrieve objects with a retractable grabber. Beyond these projects, we continue to push the limits with ultra-fast drones and rare, innovative designs.

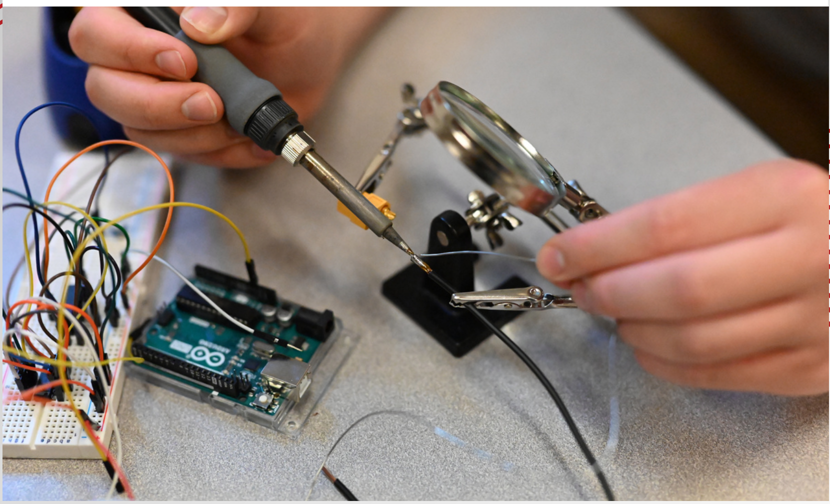
Our team of over 40 members is divided into Mechanical, Electrical, Computer Science, and Business subteams. Each contributes its unique expertise, making us not just a team of skilled engineers and strategists, but also a close-knit community of friends who share a common passion for innovation in aerial robotics.

OUR GOALS

WHAT WE DO:

Our goal is to tackle extreme, real-world challenges by designing, building, and flying advanced drones that perform beyond typical collegiate benchmarks. Through multi-semester projects, we push the state-of-the-art in autonomous flight, control, perception, and systems engineering.

Our rigorous schedule includes weekly build nights, flight tests, and cross-subteam reviews to keep projects moving quickly and safely. Beyond the technical work, we engage in regular social events that strengthen collaboration. As we develop platforms like an autonomous glider launched from our carrier drone, a gesture-controlled drone, and “Buttery” for autonomous retrieval tasks, we remain committed to innovation, teamwork, and excellence. (Our website: “an interdisciplinary student-run project team that designs, builds, and operates aerial robotics.”)



OUR SUBTEAMS

01

MECHANICAL

The Mechanical subteam owns structures and mechanisms: airframes, wings, mounts, linkages, and recovery systems. We design for high performance and serviceability, then fabricate via CNC-milled carbon fiber and aluminum, plus extensive 3D printing. The team leads assembly, integration, and test-readiness in close partnership with Electrical and CS.

02

ELECTRICAL

The Electrical subteam delivers reliable power and avionics for all platforms. We develop bus bars and harnesses, select and tune ESCs and batteries, integrate sensors and radios, and implement robust communications and safety interlocks. We prototype PCBs and own bench testing, telemetry health, and flight-ready wiring.

03

COMPUTER SCIENCE

The Computer Science subteam provides autonomy and control. We implement perception, gesture input, waypoint tracking, and mission logic; tune flight controllers; and build software that fuses sensors for reliable behavior. From obstacle navigation to autonomous retrieval, CS turns airframes into intelligent systems.

04

BUSINESS

The Business subteam manages partnerships, fundraising, branding, and outreach. We handle finances and procurement, maintain the website and media, and produce technical highlights that showcase our work to sponsors and the public.

WHY FUND US?

WE INVITE YOU TO JOIN US IN MAKING HISTORY WITH THE CORNELL UNIVERSITY AUTONOMOUS DRONE PROJECT TEAM.

OUR MISSION IS TO PUSH THE BOUNDARIES OF TECHNOLOGY AND INNOVATION BY DEVELOPING AUTONOMOUS DRONES THAT CAN CHANGE THE WORLD. HERE'S WHY YOUR SUPPORT MATTERS:

Innovation That Matters

We're not just building drones; we're shaping the future of technology. Our autonomous drones have the potential to revolutionize industries. We are engineering the impossible, for the possible of tomorrow. Your support directly fuels this innovation.

Tomorrow's Leaders

By supporting us, you're investing in the next generation of leaders, engineers, and innovators. Your contribution helps us provide invaluable hands-on experience to students, equipping them with skills and knowledge that will drive progress in the tech industry.

Real-World Impact

Our project is not just theoretical; it's hands-on and practical. We're tackling real-world challenges, and your donation will help us address these issues effectively. From enhancing agricultural practices to aiding in emergency response, our drones will make a tangible impact on society.

Your contribution, no matter the size, is a crucial part of our journey. Whether you're a tech enthusiast, an advocate for education, or simply passionate about cutting-edge innovation, we invite you to become part of our community.

Join us in shaping the future. Support the Cornell University Autonomous Drone Project Team today and be a part of the innovation that will define tomorrow. Together, we can achieve great things.

Thank you for being a part of our mission!

DONOR LEVELS

BRONZE DRONE ENTHUSIAST ($200+)

- Acknowledgement and logo on our website (cuad.info)
- Regular project updates & insights
- Resume Book access

SILVER SKY INNOVATOR ($500+)

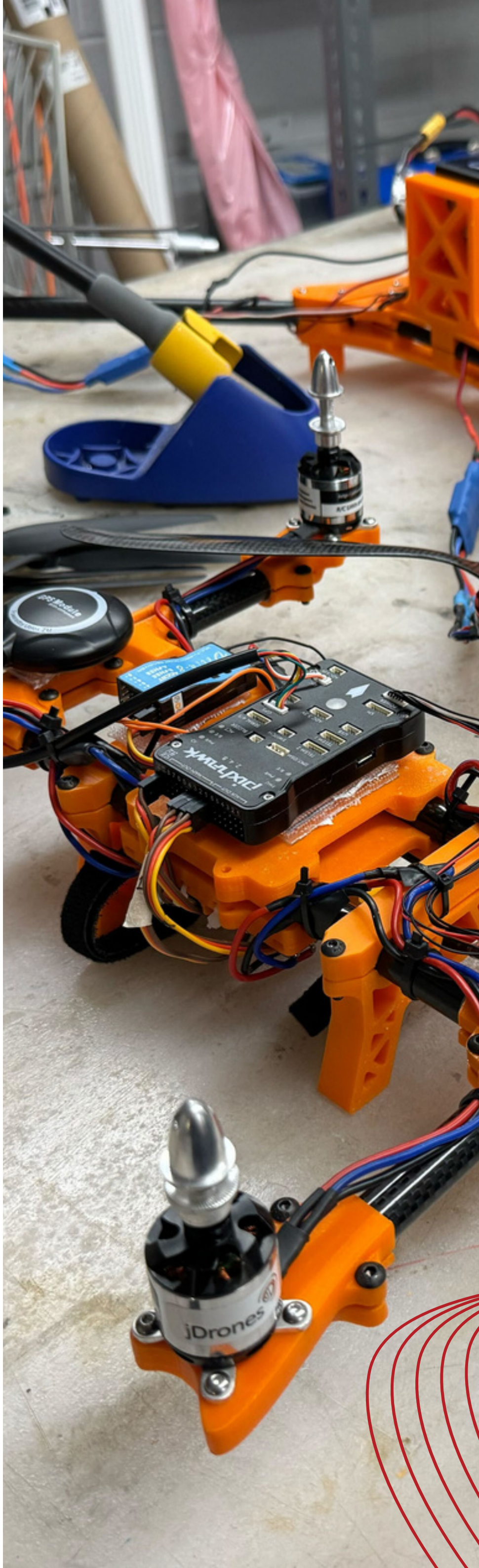
- All benefits from Drone Enthusiast
- Small logo clearly displayed on our project materials & merchandise

GOLD FLIGHT PIONEER ($1000+)

- All benefits from Sky Innovator
- Exclusive access to behind-the-scenes content, including early access to project developments.
- Medium logo on project materials & merchandise
- Company logo on the drone itself

PLATINUM AERIAL ENGINEER ($2500+)

- All benefits from Flight Pioneer
- Opportunities for direct collaboration with our team, such as joint research projects, workshops, or on-campus information sessions.
- Dedicated promotional social media announcement
- Large logo on project materials & merchandise



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